education sciences

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Mindsets Matter: A Mediation Analysis of the Role of a Technological Growth Mindset in Generative Artificial Intelligence Usage in Higher Education

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Abstract: In the digital era, generative artificial intelligence (GAI) is increasingly used in higher education, yet the psychological factors influencing its adoption are underexplored. This study examines the role of a growth mindset towards technology, defined as the belief that technological abilities can be developed in predicting GAI usage among Chinese undergraduates. Using the Unified Theory of Acceptance and Use of Technology (UTAUT), this study explored the mediating roles of performance expectancy, effort expectancy, and technology anxiety. A total of 500 students participated in an online survey. Mediation analysis showed that a growth mindset predicted GAI usage through performance expectancy, effort expectancy, and technology anxiety, even when perceived external resources and gender were statistically controlled. The findings underscore the importance of psychological readiness, alongside technical skills, in fostering GAI adoption in education. Future research should use longitudinal and experimental designs to validate these results.

Keywords: generative artificial intelligence; technology adoption; growth mindset; technology anxiety; performance expectancy; effort expectancy; higher education

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Academic Editors: Aleksandra Klašnja-Milićević, Boban Vesin and Dunja Vrbaški

Received: 14 December 2024
Revised: 4 February 2025
Accepted: 23 February 2025
Published: 3 March 2025

Citation: Chow, T. S., & To, K. (2025). Mindsets Matter: A Mediation Analysis of the Role of a Technological Growth Mindset in Generative Artificial Intelligence Usage in Higher Education. *Education Sciences*, *15*(3), 310. https://doi.org/10.3390/educsci15030310

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1. Introduction

In the current digital era, the integration of technology into educational practices has become increasingly prevalent. One impactful technological advancement that shows great promise for transforming teaching and learning is generative artificial intelligence (GAI), a subset of AI that utilizes algorithms to autonomously generate new content, such as text, images, or other data, based on patterns learned from existing datasets (Epstein et al., 2023). As scholars recently suggested, GAI can be defined as "a technology that (i) leverages deep learning models to (ii) generate human-like content (e.g., images, words) in response to (iii) complex and varied prompts (e.g., languages, instructions, questions)" (Lim et al., 2023, p. 2).

Since its release in November 2022, ChatGPT has amassed over 200 million monthly users (Babu, 2024), meanwhile sparking debates on its educational implications (Baidoo-Anu & Ansah, 2023; Lim et al., 2023; Su & Yang, 2023). While GAI offers benefits such as personalized learning, automated grading, and interactive environments, it also raises concerns about academic integrity, data privacy, algorithmic bias, and the reliability of assessments (Stokel-Walker, 2022; Gupta et al., 2023; Whalen & Mouza, 2023; Farrelly & Baker, 2023). Despite these challenges, the rapid adoption of GAI highlights the need for higher education to equip students and educators with AI literacy, as emphasized in

the DigComp 2.2 framework (Vuorikari et al., 2022). The challenges and opportunities in preparing students for this new reality are vast, necessitating a collective effort from teachers, students, researchers, and universities' management teams.

The critical importance of understanding the determinants influencing the behavioral intention to adopt GAI in teaching and learning is underscored by insights gleaned from previous studies on other learning technologies. For instance, previous studies found that perceived usefulness and perceived ease of use predicted university students' tendency to adopt cloud-computing tools in learning, which in turn predicted academic performance (Ali et al., 2018). Another study also revealed that attitudes towards e-learning predicted the adoption of e-learning in non-technology-intensive courses (Buche et al., 2012). The transformative potential of GAI in education necessitates a nuanced exploration of students' receptivity in higher education. As teachers and researchers, it is important to uncover the intricate interplay of factors influencing the intention to learn, understand, adopt, and explore creative uses of this technology.

Moreover, existing research surrounding the incorporation of technologies in education has consistently highlighted the need to align pedagogical practices with students' attributes, motivations, and intentions (Truong, 2016; Weiser et al., 2018; Yilmaz, 2017). This alignment ensures that technology serves as a facilitator rather than a disruptor of educational processes. While external factors like institutional support and available resources can influence technology adoption, it is equally crucial to understand the internal psychological factors that drive students' engagement with GAI. This study examined how these psychological factors impacted GAI adoption above and beyond the perceived availability of external resources.

1.1. Students' Mindset of Technology: Is Technological Capacity Malleable?

Accumulating research has underscored the significance of students' individual attributes, such as socioeconomic background, personality traits, and attitudes, in explaining technology adoption (Maican et al., 2019; Rivers, 2021). For instance, a study among Korean undergraduates during the pandemic found that personal innovativeness moderated the effects of subjective norms on students' technology adoption (Kim et al., 2021). Another study in Canada revealed that personality traits, such as the Big Five personality traits, have varying impacts on the intention to adopt desktop video conferencing technology in an information management course among university students (Lakhal & Khechine, 2017).

Among these individual factors, one potentially significant yet relatively under-researched aspect is students' implicit theory of technology. Students may hold varying implicit theories of technology, reflecting their beliefs about the malleability of their technological skills and abilities. People who endorse an incremental theory (i.e., a growth mindset) of technology believe that technological capacity is determined by learning and effort, while those who endorse an entity theory (i.e., fixed mindset) believe that technological capacity is a fixed construct that is difficult to develop and change (Pybus & Gillan, 2015). This concept draws from extensive research on implicit theories of intelligence, which refers to belief systems that categorize intelligence as either fixed or changeable (Dweck & Leggett, 1988). The fixed mindset is associated with a fear of failure, less persistence in learning new skills, performance anxiety, and a tendency to adopt behaviors that hinder self-regulatory strategies in problem-solving (Dweck, 1999; Stipek & Gralinski, 1996). Conversely, individuals holding a growth mindset respond more proactively to challenges, viewing them as opportunities for learning and growth. They persist longer on challenging tasks and experience more positive emotions during learning.

Recent research differentiates between general theories of intelligence and domain-specific implicit theories (Karlen & Hertel, 2021). Domain-general implicit theories of

intelligence concern the malleability of one's core intelligence that can be generalized across subject areas. Nevertheless, emerging research suggests that domain-specific implicit theories differ from general implicit theories of intelligence and might better predict learning behaviors and outcomes in specific subjects. Domain-specific implicit theories refer to beliefs about whether particular abilities, such as mathematics, technology, or literacy, are malleable. Domain-specific implicit theories capture contextually relevant information when evaluating the malleability of abilities in certain areas. Pybus and Gillan (2015) found that participants who believed that technological abilities are malleable performed better in a technology task that required identifying specific elements on a website screenshot than those who believed that technological abilities are relatively fixed. Recent findings by Dinh (2024) also showed that a growth mindset positively predicted students' attitudes and intentions towards using ChatGPT. Despite increasing interest and research on the role of individuals' characteristics in technology adoption, surprisingly little is known about the influence of technology-specific implicit theories and the underlying mechanisms. This current study sought to elucidate whether and how a growth mindset of technology affects the adoption of GAI tools in learning in higher education.

1.2. Understanding the Impacts of Growth Mindset of Technology via the Unified Theory of Acceptance and Use of Technology

In the context of the Unified Theory of Acceptance and Use of Technology (UTAUT), the adoption of technology is determined by one's intention to use it and four factors that can shape the intention, including (1) performance expectancy, (2) effort expectancy, (3) social influence, and (4) facilitating conditions (Venkatesh et al., 2003). The validity of the UTAUT model in explaining technology adoption has been supported by meta-analytic reviews (Faaeq et al., 2013; Khan et al., 2022). Among these factors, the two expectancy beliefs are pivotal in explaining the beneficial effects of a growth mindset of technology.

Performance expectancy refers to the extent to which an individual believes that utilizing a particular technology will enhance their job or learning performance (Venkatesh et al., 2003). It encompasses the overall evaluation of the perceived usefulness, outcome expectations, and relative advantage of the new technology over other tools, each contributing to a user's overall belief in the technology's potential to improve their task performance.

Effort expectancy is akin to perceived ease of use and pertains to the user's perception of the effort required to utilize the technology effectively (Venkatesh et al., 2003). The fundamental aspects of effort expectancy include perceived ease of use, perceived complexity, and user-friendliness, all of which contribute to the user's overall assessment of the effort required to learn and use the technology (Venkatesh et al., 2003). When users perceive a technology as easy to use, with minimal complexity and high user-friendliness, they are more inclined to adopt and continue using it.

Research in higher education has supported the critical roles of effort expectancy and performance expectancy in students' use of various educational technologies, including classroom response systems (Decman, 2020; Wong et al., 2019), virtual reality technology (Abd Majid & Mohd Shamsudin, 2019; Sagnier et al., 2020), and subject-specific e-learning platforms (Natasia et al., 2022) in educational settings. For example, Jang et al. (2021) found that technological, pedagogical, and content knowledge (TPACK); social norms; and motivational support predicted teachers' intention to use VR and AR in teaching by increasing performance expectancy and effort expectancy.

A growth mindset elicits self-regulated learning behaviors—such as seeking feedback, pursuing additional information, and persisting despite initial setbacks—that in turn enhance performance expectancy (Burnette et al., 2013; Orosz et al., 2023; Peng & Tullis, 2020). When applied to GAI utilization, students with a growth mindset toward technology may be more inclined to actively ask questions and seek feedback on their usage. Consequently,

they are better poised to grasp the utility of these tools. This continuous cycle of engagement and improvement reinforces students' confidence in their ability to succeed with GAI, thereby boosting their performance expectancy.

Beyond enhancing performance expectancy, a growth mindset toward technology may also diminish the perceived difficulty and complexity associated with using GAI, thereby reducing effort expectancy. Studies indicate that individuals with a growth mindset are more likely to embrace challenges as opportunities for growth and learning (Dweck, 1999). A growth mindset encourages individuals to view technological challenges as opportunities for learning and growth. It helps them perceive these challenges positively rather than as insurmountable hurdles. Research suggests that individuals with a growth mindset are inclined to seek help and utilize available resources effectively (Blackwell et al., 2007). In a technological context, this translates to users leveraging tutorials, online resources, and training programs to facilitate their learning process, thereby decreasing overall effort expectancy. According to the UTAUT, the changes in performance and effort expectancy may explain how a growth mindset of technology increases the adoption of GAI in learning.

The UTAUT also highlighted the critical role of "social influence" and "facilitating conditions", such as institutional policies and resource availability, in shaping technology usage (Venkatesh et al., 2003) and how they might influence students' intention of AI adoption. Policies governing the use of GAI in educational settings can either encourage or hinder adoption, depending on their level of restrictiveness or supportiveness (Farrelly & Baker, 2023; Whalen & Mouza, 2023). Similarly, the availability of resources, such as workshops, guides, and subscriptions to GAI tools, can enhance students' access to and comfort with these technologies (Almaiah et al., 2022; Vuorikari et al., 2022; F. Wang et al., 2023). While these social and contextual factors are undeniably important and supported by prior research (Lim et al., 2023), this present study focused on examining the psychological determinants of GAI adoption above and beyond perceived resources.

1.3. Growth Mindset and Technology Anxiety

Technology anxiety, defined as a negative emotional response to the use of technology, is characterized by apprehension and tension stemming from anticipated negative outcomes associated with technology use (Wilson et al., 2023). Initially referred to as computer anxiety in the 1970s, technology anxiety has been a subject of research since that time (Cambre & Cook, 1985). As information and communication technologies (ICT) have become more prevalent in various aspects of life, the understanding of technology anxiety has broadened to encompass a range of digital technologies beyond traditional computers.

Research indicates that technology anxiety can lead to technology avoidance, where individuals may avoid using certain technologies, thereby hindering their ability to engage with essential tools for communication, education, and work (Dönmez-Turan & Kır, 2019; Meuter et al., 2003). In educational settings, technology anxiety can impede learning opportunities, as students may be less willing to engage with digital resources or participate in technology-driven activities (Almaiah et al., 2022; Jahromi et al., 2016). A recent study suggested that technology anxiety is associated with academic burnout in online classes (Churampi-Cangalaya et al., 2024). When learning to use new technologies, such as GAI, students with high levels of technology anxiety may find these technologies intimidating, leading to avoidance behaviors. Beyond performance anxiety, the use of GAI may also elicit ethical anxiety, stemming from concerns and apprehensions about the ethical controversies surrounding AI usage (Zhu et al., 2024).

We propose that a growth mindset towards technology can significantly reduce technology anxiety. Technology anxiety often stems from the anticipation of negative outcomes, such as failure or embarrassment when using new technology (Beckers & Schmidt, 2003).

A growth mindset could mitigate this anticipation because individuals understand that mistakes are part of the learning process and not indicative of their overall capability. This understanding helps alleviate the stress and apprehension that contribute to technology anxiety. Support-seeking behaviors driven by a growth mindset could also help to alleviate anxiety (Burnette et al., 2013). Extensive research demonstrated that a growth mindset can reduce anxiety and stress while learning novel and challenging tasks (King et al., 2012; Jiang et al., 2024). In a recent large-scale intervention study that involved over two thousand adolescents and undergraduates, Yeager et al. (2022) showed that a growth mindset intervention reduced both physiological indicators and self-reported stress and anxiety. This present study proposes that a growth mindset towards technology reduces technology anxiety, which in turn increases GAI usage in learning.

Taken together, our study aims to explore the relationship between the growth mindset of technology and GAI usage. Drawing on the UTAUT theory and research, we further postulate that the relationship is mediated by performance expectancy, effort expectancy, and technology anxiety, as the following hypotheses indicate:

H1: *A growth mindset of technology is positively related to GAI usage behaviors.*

H2: *Performance expectancy mediates the relationship between a growth mindset of technology and GAI usage behaviors.*

H3: *Effort expectancy mediates the relationship between a growth mindset of technology and GAI usage behaviors.*

H4: *Technology anxiety mediates the relationship between the growth mindset of technology and GAI usage behaviors.*

2. Materials and Methods

2.1. Participants and Procedure

An a priori power analysis was conducted to determine the minimum sample size required for the analysis, which included seven variables: one independent variable, one dependent variable, three mediators, and two covariates. The analysis was based on a multiple regression model with a medium effect size ($f^2 = 0.15$), a significance level of $\alpha = 0.05$, and a desired power of 0.80. We used G*Power 3.1 (Faul et al., 2009) to perform the calculation. The results indicated that a minimum of 103 participants was required for a regression model with seven predictors. However, because the study involved testing mediation effects, which typically require larger sample sizes to achieve adequate statistical power, we followed Fritz and MacKinnon's (2007) recommendation for detecting mediation in models with medium effect sizes. For a three-mediator model, they suggest a minimum sample size of 462 participants to achieve 0.80 power. To ensure robust statistical power, we targeted to recruit a total of 500 participants to ensure sufficient power to detect meaningful effects.

A total of 500 Chinese university undergraduates were recruited as participants for this present study after it was approved by the Human Research Ethics Committee (HREC) of Hong Kong Shue Yan University under Reference No. HREC 24-3(2). The recruitment was conducted through an online survey platform called Credamo, which is widely used in China and contains a user base of over three million individuals. Prior to commencing the questionnaire, informed consent was obtained from all participants. Participants were instructed that all questions needed to be filled out before they could successfully submit the questionnaire. Participants were then invited to fill out an online survey, which took around 20 minutes to complete. The participants had a mean age of 21.79 years (SD = 1.56).

In our sample, there were 359 females (71.8%), 140 males (28.0%), and one who categorized their gender as other (0.2%). For further details about the participants, please refer to Table 1.

Table 1. Demographics of the participants.

	Percentage (*N* = 500)
Year of Study	
Year 1	6.2
Year 2	16.0
Year 3	30.1
Year 4	34.7
Year 5 or above	13.0
Major of Study	
Business Administration	22.8
Science and Medicine	21.0
Engineering and Architecture	20.0
Arts and Law	14.4
Social Sciences	8.4
Education	7.6
Others	5.6
Yearly Family Income	
less than 3000 RMB	4.8
3001 to 6000 RMB	14.6
6001 to 9000 RMB	18
9001 to 12,000 RMB	30.2
12,001 to 15,000 RMB	11.2
15,001 to 18,000 RMB	9.6
more than 18,000 RMB	11.6

2.2. Measurements

The following are the scales that comprised the survey. To assess participants whose first language is Chinese, the scales were translated from validated English versions and adapted to fit the context of this study on GAI. To ensure validity, two authors, who are both fluent in English and Chinese, conducted two rounds of back-and-forth translation. A pilot test was then conducted with ten university students from the authors' institution who were pursuing degree programs and whose first language is Chinese for revision of wordings. These students were provided with bilingual items and asked to evaluate the consistency of their verbal meaning. For analyses, mean scores were calculated for each scale. The survey items used in this study can be found in Appendix A.

2.2.1. GAI Usage Behavior

The GAI Behavioral Usage scale used in this study comprised four items, adapted from the usage subscale of the UTAUT scale developed by Abbad (2021). The wording of the items was modified to specifically assess the extent to which participants utilized GAI. A sample item of the scale is "I do most learning tasks by using GAI". Participants were instructed to rate each item on a 7-point Likert scale, ranging from "strongly disagree" to "strongly agree". The internal consistency of the GAI Behavioral Usage scale in this study was found to be good, with a Cronbach's alpha coefficient of 0.81.

2.2.2. Growth Mindset of Technology

The measurement of the growth mindset of technology was adapted from Pybus and Gillan (2015)'s scale. It consisted of five items that assessed participants' perspectives on whether their technology-related skills could be developed by learning and effort. A

sample item from this scale demonstrating the growth mindset is "your technology-related skills are something that you can develop". On the other hand, a sample item reflecting fixed mindset is "your technology ability is something about you that you can't change very much". Participants were requested to rate each item on a 7-point Likert scale, ranging from "strongly disagree" to "strongly agree". During the analysis, two items that reflected a fixed mindset were reverse-coded so that higher scores on the scale would indicate a stronger growth mindset. In this study, the internal consistency of the implicit theories of technology scale was found to be satisfactory, with a Cronbach's alpha coefficient of 0.74.

2.2.3. Performance Expectancy

The four-item performance expectancy scale used in this study was derived and adapted from the original performance expectancy subscale of the UTAUT scale developed by Abbad (2021). The scale was specifically modified to assess participants' perception of the extent to which using GAI would enhance their productivity and effectiveness in achieving their goals. A sample item of this scale is "using GAI increases my learning productivity". Participants were provided with instructions to rate each item on a 7-point Likert scale, ranging from "strongly disagree" to "strongly agree". The performance expectancy scale demonstrated acceptable internal consistency in this study, with a Cronbach's alpha coefficient of 0.71.

2.2.4. Effort Expectancy

The four-item effort expectancy scale was derived and adapted from the effort expectancy subscale of the UTAUT scale developed by Abbad (2021). The scale was specifically modified to assess participants' perception of the ease of using GAI. A sample item of this scale is "learning to use GAI is easy for me". Participants were instructed to rate each item on a 7-point Likert scale, ranging from "strongly disagree" to "strongly agree". The effort expectancy scale demonstrated good internal consistency in this study, with a Cronbach's alpha coefficient of 0.87.

2.2.5. Technology Anxiety

We utilized the 11-item scale developed by Wilson et al. (2023) to assess students' technology anxiety. The scale measures the extent of anxiety individuals experience regarding the use of technology, with higher scores indicating a greater level of anxiety. Participants were provided with statements such as "using technology makes me nervous". and asked to rate their agreement on a 7-point Likert scale, spanning from "strongly disagree" to "strongly agree". In this study, the technology anxiety scale demonstrated good internal consistency, as indicated by a Cronbach's alpha coefficient of 0.83.

2.2.6. Perceived Resources

Perceived resources for supporting GAI usage were included as a covariate in our analysis to examine the predictive value of a growth mindset above and beyond external supportive factors. The measurement of perceived resources included four items adapted from the perceived resources and technology acceptance model developed by Sivo et al. (2018). An example item from this scale is "I have access to the resources I would need to use GAI in the courses". Participants rated these items on a 7-point Likert scale, ranging from "extremely unlikely" to "extremely likely". The perceived resources scale exhibited high internal consistency in this study, with a Cronbach's alpha coefficient of 0.84.

2.3. Data Analysis

The data analysis was performed using the statistical software package SPSS 26.0. Prior to hypothesis testing, descriptive analysis was conducted to provide an overview

of the participants' demographics and characteristics. Pearson's correlation analysis was employed to examine the relationships among variables.

To test the hypothesis, a parallel mediation analysis was carried out to examine our proposed mediating model. The total, direct, and indirect effects were estimated using the SPSS PROCESS macro (model 4). The dependent variable (DV) in this model was identified as GAI usage behavior, while the independent variable (IV) was the growth mindset of technology. Effort expectancy (MV1), performance expectancy (MV2), and technology anxiety (MV3) were proposed as mediators between growth mindset and GAI usage behavior. Perceived resources for supporting GAI usage were included as a covariate in our analysis to examine the predictive value of a growth mindset above and beyond external supportive factors. This allows us to ascertain whether the psychological factors influence GAI adoption independently of the available resources. Specifically, we calculated 95% confidence intervals for the indirect effects using percentile bootstrap estimates with 5000 iterations (Hayes, 2017).

3. Results

3.1. Preliminary Analysis

The results of descriptive statistics and correlations are presented in Table 2. All variables used in this study are significantly correlated ($p < 0.001$). Specifically, GAI usage behaviors, growth mindset of technology, performance expectancy, and effort expectancy are positively correlated with each other, while technology anxiety is negatively correlated with other variables. As hypothesized, the growth mindset of technology was positively correlated with the GAI usage behavior among Chinese undergraduates. Since our data were skewed toward a higher representation of females, an analysis of gender differences in the main variables was conducted to determine whether gender should be included as a covariate in the final model. Significant gender differences were found in technology anxiety ($t = 5.83$, $p < 0.001$), performance expectancy ($t = -5.31$, $p < 0.001$), effort expectancy ($t = -3.88$, $p < 0.001$), growth mindset of technology ($t = -4.90$, $p < 0.001$), and GAI usage ($t = -5.25$, $p < 0.001$), with female participants found to have higher technology anxiety and lower performance expectancy, effort expectancy, and growth mindset of technology, as well as GAI usage, than male participants. In light of this result, gender was also included as a covariate in our mediation analysis.

Participants' frequency of using GAI, purposes of using GAI, and their universities' policies and support are shown in Tables 3 and 4. The findings from Table 3 provide a clearer understanding of GAI usage within the sample. Notably, 98.8% of participants reported using GAI at least two to three times per month during the last semester. The most common uses of GAI were searching for information (74.8%), writing (64.0%), organizing learning (63.6%), and understanding learning materials (62.0%), with more than half of the participants reporting engagement in these activities. On the other hand, the findings from Table 4 paint a varied picture of universities' resources and policies regarding GAI usage. While over a third of participants (34.8%) indicated that their university provided no resources to support GAI usage, nearly half (48.0%) reported access to guides or practical examples. Additionally, 30.8% mentioned newsletters or promotions issued by their universities, and 32.2% noted announcements for subscribing to GAI tools as forms of institutional support.

Table 2. Descriptive statistics and correlations between study variables.

Variable	1	2	3	4	5	6
1. GAI usage behavior	1					
2. Growth mindset	0.30 ***	1				
3. Performance expectancy	0.59 ***	0.35 ***	1			
4. Effort expectancy	0.52 ***	0.43 ***	0.48 ***	1		
5. Technology anxiety	−0.50 ***	−0.51 ***	0.47 ***	−0.52 ***	1	
6. Perceived resource	0.57 ***	0.39 ***	0.56 ***	0.62 ***	−0.49 ***	1
Mean	5.09	5.29	5.67	5.46	2.13	4.96
SD	1.07	0.84	0.70	0.94	0.55	1.09
Cronbach's α	0.81	0.74	0.71	0.87	0.83	0.84

*** $p < 0.001$.

Table 3. Participants' usage of GAI in this study.

	Percentage (*N* = 500)
Frequency of GAI Usage in Past Semester	
More than once every day	4.4
Everyday	15.4
Once a week	9.4
Two to three times per week	34.1
Once a month	10.8
Two to three times per month	24.6
Never	1.2
Purpose of Using GAI in Study	
Searching Information	74.8
Writing	64.0
Organizing learning	63.6
Understanding learning materials	62.0
Taking notes	20.2
Designing graphics	19.0
Creating presentations	15.8
Converting audio to text	14.0
Brainstorming	13.6
Checking for plagiarism	9.4
Creating videos	3.2
Creating personal mentorship	2.8
Other	2.4
Creating sound effects	2.0

Table 4. Participants' university policy and resources on GAI.

	Percentage (*N* = 500)
University's Policy on the use of GAI	
No restrictions	20.8
Allowed for specific learning tasks or subjects	27.8
Allow instructors to make individual decisions	16.0
Ambiguous	23.4
Prohibited for all learning tasks	3.6
Do not know	8.4

Table 4. *Cont.*

	Percentage (*N* = 500)
University's Resource on the use of GAI	
Nil	34.8
Workshops	7.4
Guides or practical examples (print/web/video)	48.0
Newsletters or promotions issued by universities	30.8
Announcements for subscribing to GAI tools	32.2
Staff for consultation or support	10.2
Other	1.8

3.2. Mediation Model

The mediation analysis results supported the hypothesized model, with significant indirect effects of a growth mindset on GAI usage behavior through performance expectancy, effort expectancy, and technology anxiety, after controlling for perceived resources and gender. This indicates that the influence of a growth mindset on GAI adoption is robust and not merely a byproduct of external resource availability; F (3, 495) = 83.29 and $p < 0.001$, with an R^2 value of 0.34, indicating that the model explains 34% of the variance in GAI usage behavior (see Figure 1). All the hypothesized specific indirect effects were found to be statistically significant (see Table 5). Specifically, the effect of a growth mindset on GAI usage behavior through performance expectancy (indirect effect = 0.06, 95% CI = 0.03, 0.10), effort expectancy (indirect effect = 0.04, 95% CI = 0.01, 0.08), and technology anxiety (indirect effect = 0.09, 95% CI = 0.05, 0.14) were statistically significant after controlling for gender and perceived resources. The total indirect effect was also statistically significant (indirect effect = 0.19, 95% CI = 0.13, 0.26).

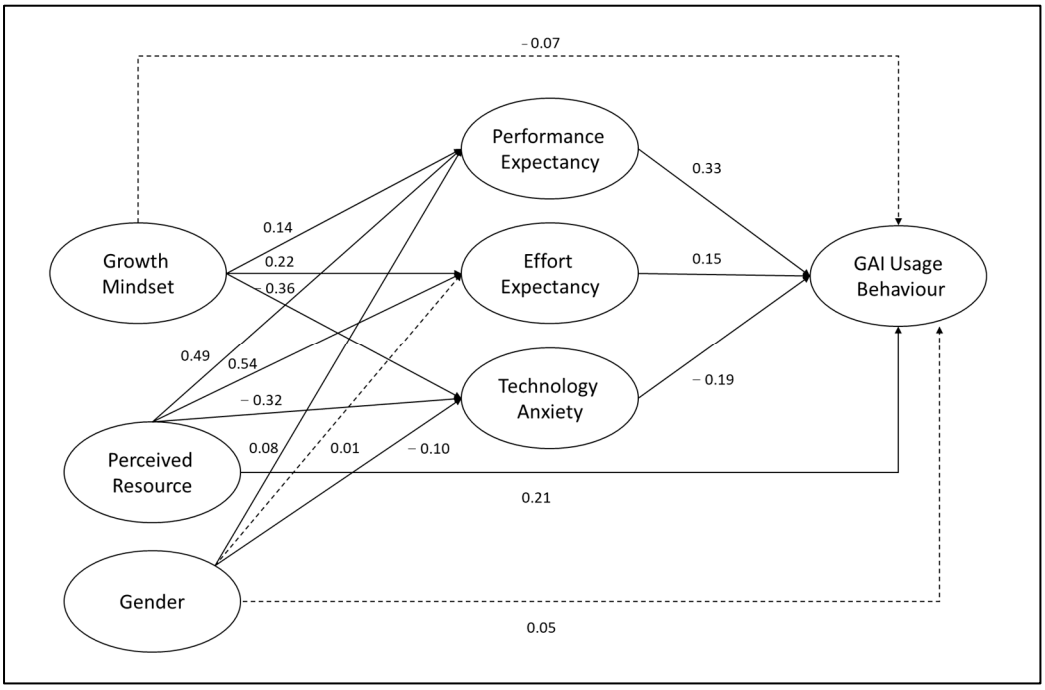

Figure 1. A path diagram illustrating the effect of growth mindset on GAI usage behavior mediated by performance expectancy, effort expectancy, and technology anxiety, controlled for perceived resources and gender. The values shown in the figure are standardized path coefficients. Solid lines represent significant paths in the hypothesized model (all *p*s < 0.05). The broken line indicates a non-significant path in the hypothesized model ($p > 0.05$). Gender is represented by a binary coding scheme, where 0 corresponds to female and 1 corresponds to male.

Table 5. Effects of growth mindset on GAI usage behavior after controlling for perceived resources and gender.

Effect on GAI Usage Behavior	Growth Mindset		
	Effect	SE	95% CI
Total Effect	0.10	0.05	[0.00, 0.20]
Direct Effect	−0.09	0.05	[−0.19, 0.00]
Total Indirect Effects	0.19	0.03	[0.13, 0.26]
Mediation via			
Performance Expectancy	0.06	0.03	[0.03, 0.10]
Effort Expectancy	0.04	0.02	[0.01, 0.08]
Technology anxiety	0.09	0.02	[0.05, 0.14]

4. Discussion

This study investigated the predictive value of a growth mindset on the usage of GAI among Chinese undergraduates. With the increasing prevalence of GAI tools, understanding the psychological and behavioral factors that impact their adoption is critical. Preliminary evidence has suggested that a growth mindset of technology, that is, the belief that one's technology capacity is malleable, predicts the adoption of new technology (Dinh, 2024; Pybus & Gillan, 2015), but the underlying mechanism remains underexplored. Our study aimed to fill these gaps by examining the indirect effects of a growth mindset of technology on GAI usage through performance expectancy, effort expectancy, and technology anxiety.

4.1. The Effects of a Growth Mindset on GAI Usage in Higher Education

The findings of this study must be interpreted within the context of students' basic experiences with GAI and the availability of institutional resources. Our findings suggested that nearly all students (98.8%) reported using GAI at least two to three times per month, engaging in tasks such as information searching, writing, and organizing learning materials. Additionally, most students reported access to practical guides, newsletters, or announcements promoting GAI tools. This context highlights a baseline level of exposure to GAI, suggesting that students in this sample had sufficient opportunities to engage with these technologies.

Our result supported H_1, which proposed that the growth mindset of technology is positively related to GAI usage behaviors. The finding is aligned with emerging research on domain-specific implicit theories, which suggest that domain-specific implicit theories predict learning behaviors and outcomes (Gunderson et al., 2017). In the context of technology learning, we found that students who believed that their technological abilities could be developed (i.e., those with a growth mindset) were more likely to engage with GAI tools. This echoes findings from previous research that indicated the importance of adaptability and growth-oriented beliefs in technology adoption (Weiser et al., 2018; Truong, 2016).

These findings reinforce the value of growth mindset interventions, particularly in the context of emerging technologies like GAI, which have been shown to require not only technical skills but also psychological readiness for effective adoption (Lakhal & Khechine, 2017). Universities and educators should focus not only on providing external resources but also on cultivating students' internal psychological readiness, as the study shows that a growth mindset towards technology could independently drive GAI usage behavior, even when external resources were controlled. Despite the widespread availability of GAI tools, a significant portion of students have yet to fully integrate these tools into their learning. This gap may contribute to a "digital divide", which could impact both curriculum design and student learning outcomes. Growth mindset interventions, such as targeted workshops

or training programs, could serve as a cost-effective means to overcome psychological barriers to technology adoption, thus encouraging effective and responsible GAI adoption, and could bridge this divide (Burnette et al., 2013).

4.2. The Role of Performance and Effort Expectancy in GAI Usage in Higher Education

Furthermore, our results highlight the mechanisms through which a growth mindset influences GAI usage behavior. We proposed that the impact of a growth mindset operates through three mediation pathways: performance expectancy, effort expectancy, and technology anxiety. The H_2 and H_3, which were developed based on the UTAUT theory, were supported by the data, indicating that performance expectancy and effort expectancy played a mediating role in the relationship between the growth mindset of technology and GAI usage. Moreover, our data also supported H_4, which suggested that technological anxiety was also a significant mediator between the growth mindset of technology and the GAI usage. Taken together, these findings suggest that students with a growth mindset perceive GAI tools as more useful and easier to use, while simultaneously experiencing lower levels of anxiety related to technology.

Our findings align with previous studies that have demonstrated the importance of perceived usefulness in technology adoption (Ali et al., 2018; Jang et al., 2021). When students believe that GAI can enhance their academic performance and productivity, they are more likely to use these tools. This echoes the findings of Jang et al. (2021), who showed that performance expectancy significantly predicted the adoption of advanced educational technologies like virtual reality.

Similarly, effort expectancy also played a significant role in mediating the relationship between growth mindset and GAI usage. Students who perceive GAI as easy to use are more inclined to adopt and engage with the technology. This finding suggests that reducing the perceived complexity of GAI through user-friendly interfaces, training, and support materials could further enhance its adoption in higher education. As Truong (2016) and Yilmaz (2017) emphasize, aligning educational technologies with student attributes, motivations, and needs is critical to ensuring that these tools serve as facilitators rather than disruptors of learning processes.

4.3. The Role of Technology Anxiety

In addition to the positive effects of performance and effort expectancy, our study also revealed the negative impact of technology anxiety on GAI usage behavior. Our findings corroborate earlier research that connects technology anxiety with decreased learning opportunities and technology avoidance (Dönmez-Turan & Kır, 2019; Meuter et al., 2003).

Although previous research has examined the detrimental effects of technology anxiety in other educational technologies (Churampi-Cangalaya et al., 2024; Almaiah et al., 2022), our study is among the first to explore its impact in the context of GAI. Our findings suggest that students who experience high levels of technology anxiety are less likely to adopt GAI. This highlights the importance of addressing emotional barriers to technology adoption, particularly in the context of emerging technologies that may be perceived as complex or intimidating (Whalen & Mouza, 2023).

The sources of technology anxiety in the context of GAI deserve further investigation. Previous research suggests that anxiety may stem from various factors, including information overload, technological complexity, and concerns about cybersecurity (Bawden & Robinson, 2009). These sources might be particularly relevant in the context of rapidly evolving tools like GAI.

Additionally, in the case of GAI, ethical concerns related to plagiarism and academic dishonesty may contribute to students' reluctance to use these tools (Stokel-Walker, 2022).

Addressing these concerns through clear guidelines and policies on GAI usage could help alleviate students' anxiety and encourage more responsible usage.

4.4. Limitations and Future Research Directions

There are several limitations of this current study. First, it is important to acknowledge the correlational nature of our study. With a cross-sectional survey, the causal effect of mindset has yet to be established in our study. Nevertheless, the existing experimental research in other domains has repeatedly demonstrated the impacts of mindsets on task perception and learning behaviors. It is also possible that there exists a bidirectional, dynamic association between students' mindsets and technology adoption. We encourage future research to explore the effects of a growth mindset through experimental or longitudinal designs. We anticipate that students who receive technology growth mindset intervention will be more inclined to perceive GAI as useful and easy to use and experience reduced anxiety, which in turn may lead to more engagement with GAI.

Second, emerging studies suggest that the notion of GAI usage may vary among different users (Lim et al., 2023; Su & Yang, 2023). In this current study, we utilized the usage behavior scale from the UTAUT as the outcome measure as we deemed it meaningful to assess the frequency of usage. However, users may also vary in effectiveness and mastery level even though they have the same usage frequency. For instance, some users may struggle to identify the most suitable GAI tools for their goals or the most effective prompts to fully leverage these tools. Future research could also explore factors that predict the effective and responsible usage of AI tools. Additionally, researchers could investigate the downstream effects of GAI usage on academic performance.

Third, this current research does not fully account for the impact of cultural differences and the technological environment on GAI usage. First, the scales used in this study were translated and adapted from tools originally developed for English-speaking communities. Although the authors implemented measures to ensure the quality of the translation, a larger-scale validation study is needed to confirm that the measurements appropriately reflect the cultural and contextual specificities of the Chinese community and the GAI context. Additionally, emerging research has started to explore the role of cultural values in technology adoption (Huang et al., 2019; Lee et al., 2013). Previous studies have also highlighted that the Chinese government's policy towards AI use emphasizes safety, which may differ from approaches taken in other countries (Gupta et al., 2023). While these studies primarily examined traditional AI rather than GAI, it is important to recognize that such factors could limit the generalizability of our findings.

Fourth, the data for this research were collected through an online survey platform, and the quality of such data should be interpreted with caution. Online platforms like MTurk have been considered significant breakthroughs in psychological and educational research over the past decade, as they enable large-scale sample collection. However, they also raise concerns about data quality (Cheung et al., 2017; Follmer et al., 2017). Drawing on the review by Cheung et al. (2017), we align with the perspective that "the quality of the data is not defined by the data source per se, but rather the decisions researchers make during the stages of study design, data collection, and data analysis" (p. 347). In this study, the survey platform Credamo was used, which is widely recognized as a reliable platform among Chinese participants. Despite its credibility, it should be noted that participants recruited via Credamo are likely to be more accustomed to participating in online research and utilizing online resources compared to the general population. Therefore, while the platform is appropriate for online research, the results of this study should be interpreted cautiously, especially when aiming to generalize findings to the broader Chinese population.

5. Conclusions

This study examined the psychological determinants of generative artificial intelligence (GAI) adoption among Chinese undergraduates, emphasizing the role of a technological growth mindset—the belief that technological abilities can be developed through effort and learning. Using the Unified Theory of Acceptance and Use of Technology (UTAUT), this research explored how performance expectancy, effort expectancy, and technology anxiety mediate the relationship between a growth mindset and GAI usage. The findings provide valuable insights into how psychological readiness complements technical skills in fostering GAI adoption in higher education.

The results confirmed that growth mindset significantly predicted GAI usage (H_1), mediated by performance expectancy (H_2), effort expectancy (H_3), and reduced technology anxiety (H_4). Students with a growth mindset viewed GAI tools as more useful, easier to use, and less anxiety-inducing, which collectively enhanced their engagement. These findings reinforce the importance of domain-specific implicit theories (Gunderson et al., 2017) and extend the UTAUT framework by integrating psychological constructs, demonstrating how mindset shapes cognitive and emotional responses to technology.

Future research could extend this exploration by adopting longitudinal or experimental designs to establish causality and assess the long-term effects of mindset interventions on GAI engagement. Cross-cultural studies are also essential to understanding how cultural values and governmental policies influence adoption patterns. On the other hand, while our study primarily focuses on the psychological factors affecting GAI usage when external support is accounted for, other social factors, such as institutional settings, subjective norms, and peer influences (Granić, 2022), could also play a significant role in technological adoption within educational contexts. Exploring how these social factors contribute to GAI usage may provide a more comprehensive understanding of how to facilitate its integration into higher education. Additionally, investigating qualitative differences in GAI usage—such as proficiency in prompt engineering or ethical applications—could further deepen insights into effective utilization.

Our results also provide some practical implications for higher education. Considering that some universities represented in our sample (34.8%) provided no resources to support the use of GAI, the growing availability of institutional resources and supportive policies could enhance the positive effects of a growth mindset by fostering environments that encourage exploration and reduce barriers to adoption. First, fostering a growth mindset through targeted interventions (e.g., workshops emphasizing effort-based learning) can reduce psychological barriers to GAI adoption. Second, institutions should ensure that GAI tools are user-friendly and offer comprehensive training programs to lower perceived effort. Third, addressing technology anxiety through supportive environments—such as peer mentoring and clear ethical guidelines—is essential, especially given its strong negative correlation with GAI usage. These strategies can bridge the "digital divide", promoting equitable access to AI-driven learning tools and empowering students to navigate an AI-augmented educational landscape.

As GAI tools like ChatGPT reshape education, this study underscores the importance of addressing both technical and psychological barriers to adoption. A growth mindset not only enhances perceived utility and ease of use but also alleviates anxiety, fostering adaptability and resilience. By prioritizing strategies that nurture these qualities, higher education institutions can empower students to thrive in an increasingly AI-driven world.

Author Contributions: Conceptualized the study, designed the methodology, and was a major contributor in writing and revising the manuscript, T.S.C.; K.T. contributed to the data curation, review and editing of the manuscript. All authors have read and agreed to the published version of the manuscript.

Funding: The work described in this paper was fully supported by a grant from the Research Grants Council of the Hong Kong Special Administrative Region, China.

Institutional Review Board Statement: The study was approved by the Human Research Ethics Committee (HREC) of Hong Kong Shue Yan University on 25 April 2024. Reference number: HREC 24-3(2).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The original data presented in the study are openly available in OSF at "osf.io/umq6f" (accessed on 4 February 2025).

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A. Items Used in This Study

Usage behavior

1. I consider myself a regular user of generative AI
2. I prefer to use generative AI when available
3. I do most learning tasks by using generative AI
4. My tendency is towards using generative AI whenever possible
5. Growth Mindset of Technology
6. You have a certain amount of technology ability, and you really can't do much to change it.
7. Your technology ability is something about you that you can't change very much.
8. Practice, hard work, effort, and persistence can change your ability to use technology.
9. No matter who you are, you can significantly change your intelligence level
10. Your technology-related skills are something that you can develop.

Performance Expectancy

1. I find generative AI useful in my learning
2. Using generative AI enables me to accomplish learning activities more quickly
3. Using generative AI increases my learning productivity
4. If I use generative AI, I will increase my chances of getting a better mark in the courses

Effort expectancy

1. My interaction with generative AI is clear and understandable
2. I am skilful at using generative AI
3. Learning to use generative AI is easy for me
4. I find it easy to get generative AI to do what I want it to do

Technology anxiety

1. I am not a technology person
2. I am reluctant to learn new features of technology
3. I am uncomfortable using technology
4. Technology does not improve my quality of life
5. I feel out of control using technology
6. I feel uneasy using technology
7. I feel technology complicates simple tasks
8. Keeping up with the newest technology is impossible
9. I am inefficient with technology

10. Using technology makes me nervous
11. I am often annoyed when using technology

Perceived resources

1. I have the resources I would need to use generative AI in the courses.
2. There are no barriers to my using generative AI in the courses.
3. I would be able to use generative AI in the course if I wanted to
4. I have access to the resources I would need to use generative AI in the courses.

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